

2 Fig. 2.1 shows the variation with displacement d of the force F applied to an object at rest.

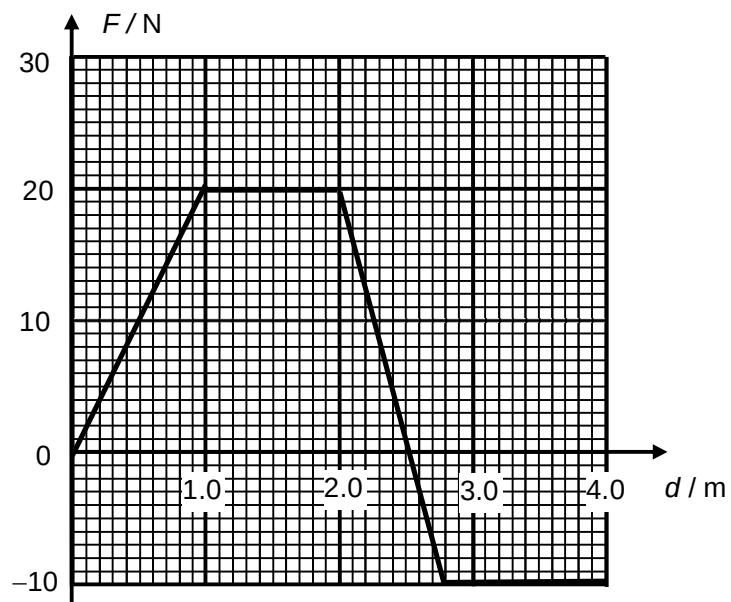


Fig. 2.1

(a) On Fig. 2.2, draw a graph showing the variation with d of the work done.

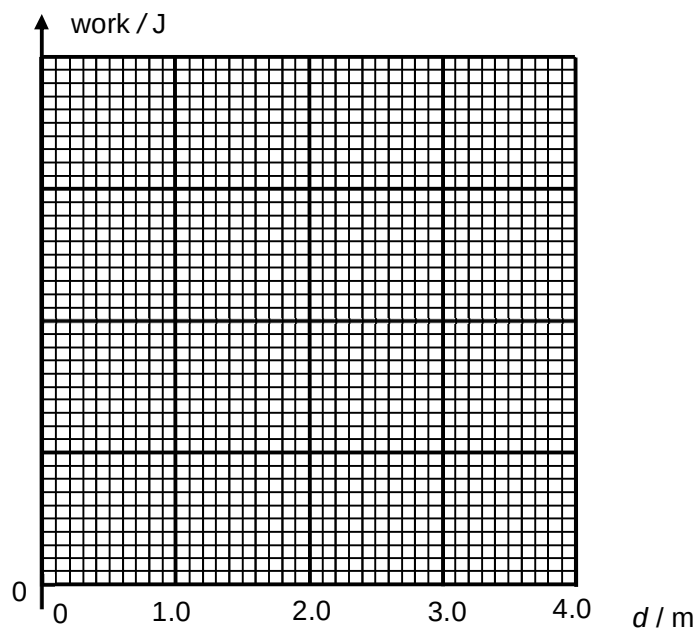


Fig. 2.2

[4]

- (b) Fig. 2.3 shows the object moving from rest up a slope of 30° as a result of the force F applied in Fig. 2.1. A constant frictional force of 3.0 N acts on the object and the object has a mass of 0.50 kg .

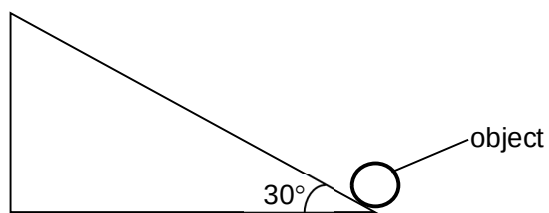


Fig. 2.3

For a displacement d of 2.0 m up the slope, determine for the object

- (i) its gain in gravitational potential energy,

gain = J [1]

- (ii) its final kinetic energy.

kinetic energy = J [3]

- 3** A bob of mass 100 g is tied to a brass weight of mass 400 g using an inelastic string